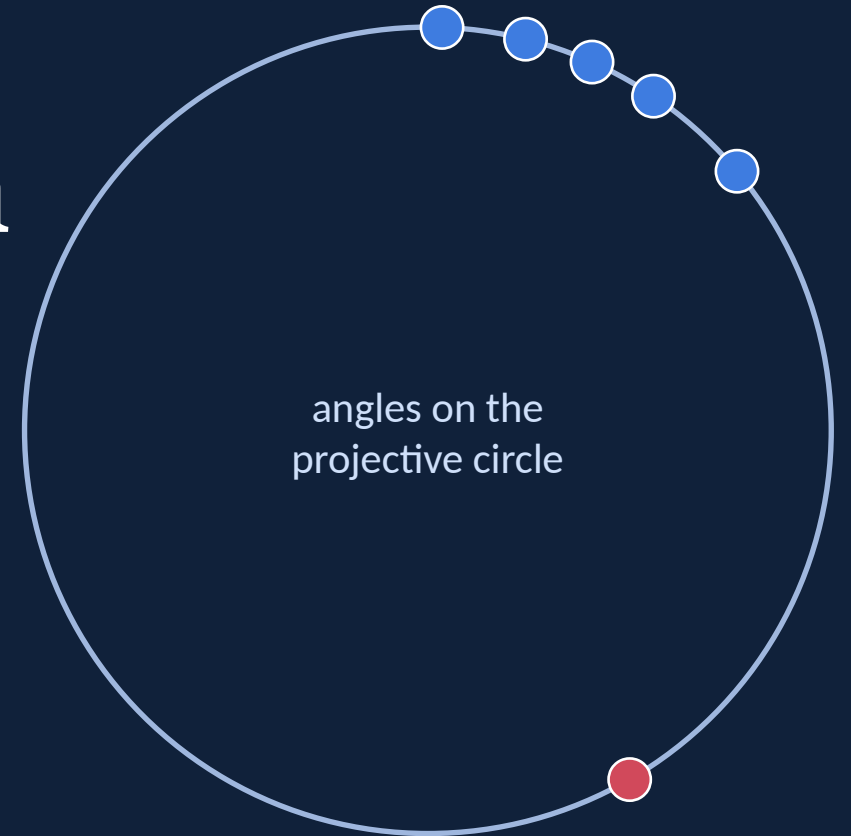


R A S h R

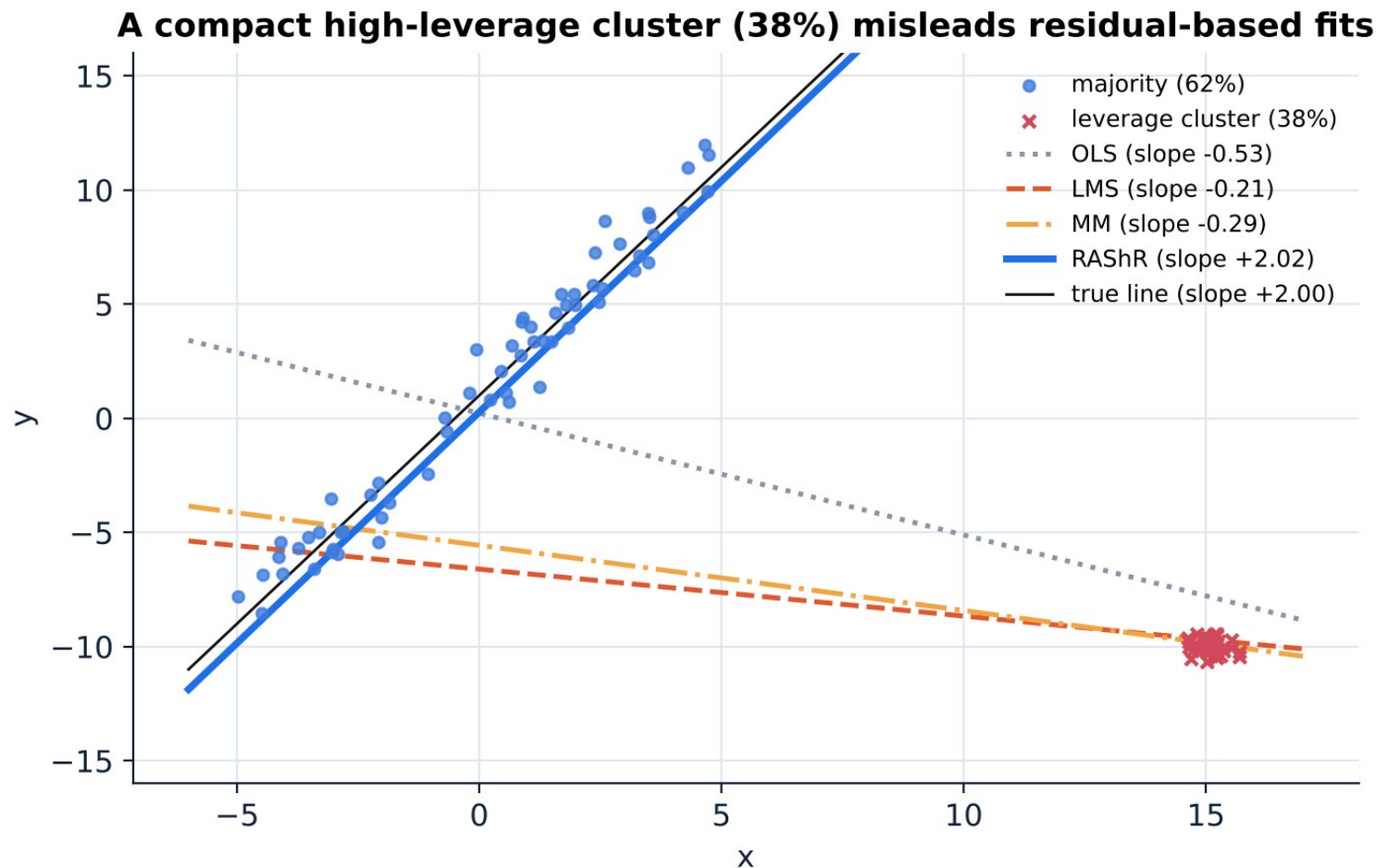
Repeated Angular Shorth Regression

Fit a line by asking which direction pairwise chords agree on, not which line has small residuals.



Based on "Robust Linear Regression via the Repeated Angular Shorth" — V. Dharavath & V. Srivastava, NIT Warangal

A robust fit can still follow the wrong line



+2.02

RAShR slope (paper)

-0.20

LMS slope (paper)

-0.26

MM slope (paper)

-0.53

OLS slope (paper)

Why surprising?

High breakdown means the estimate stays bounded — not that it follows the dominant structure.

Our re-creation; paper's exact sample not available.

Vocabulary, built up step by step

Outlier vs contamination

Outlier: one unusual point.

Contamination: observations not following the main pattern — possibly many.

Coherent contamination

Contaminated points related to each other: a compact cluster or a competing line.

High leverage

x far from the bulk of the x-values → strong pull on the slope.

Breakdown point

Fraction that can be replaced before the estimate can be driven arbitrarily far: a boundedness guarantee.

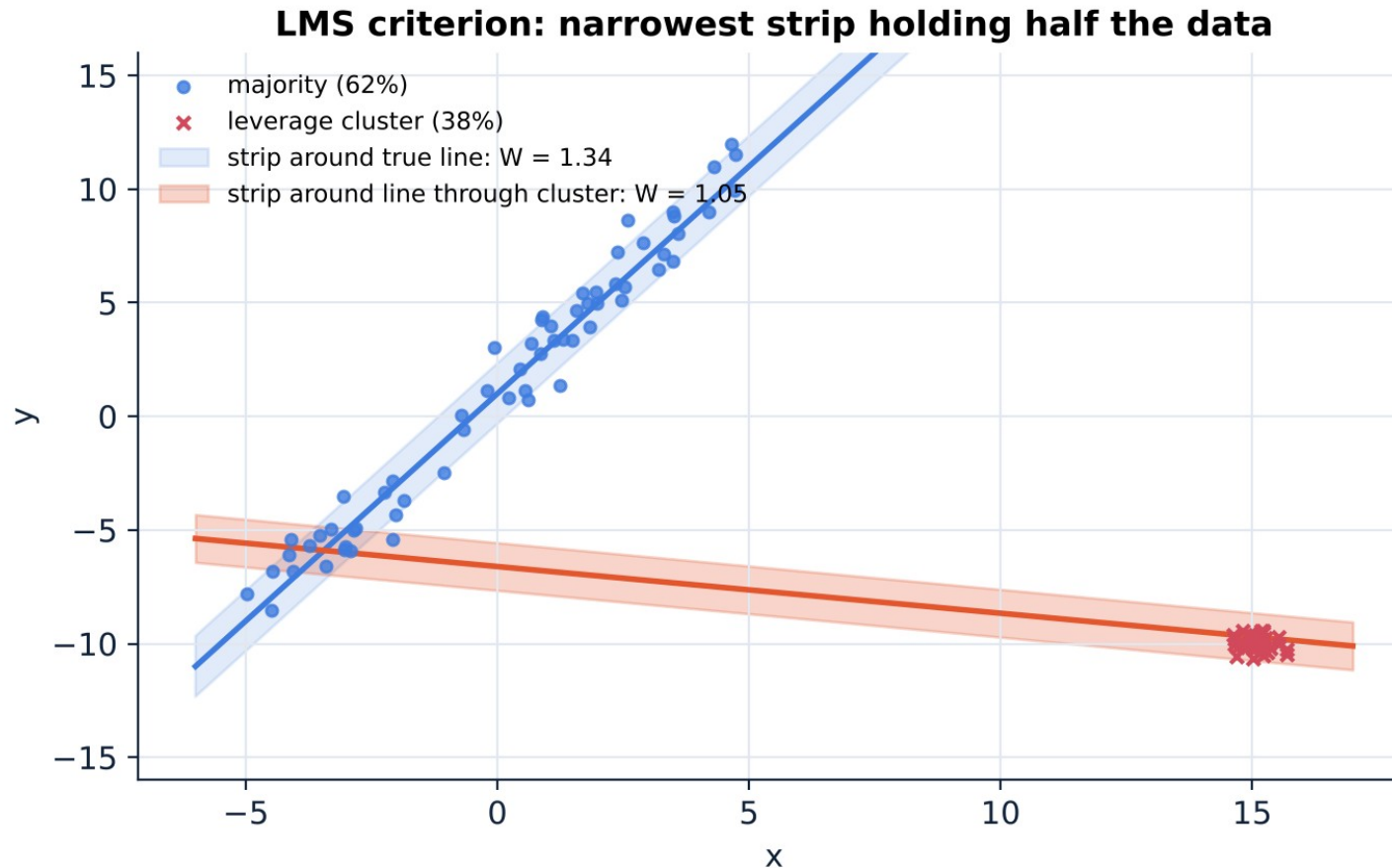
Residual-based method

Judges candidate lines by residuals $y_i - \hat{y}_i$:
LMS, LTS, MM...

Chord

The segment joining two points. It has a direction; it is not a regression line.

Why residuals mislead: a narrow strip through the cluster



$$W(\ell) = \inf \{ w : \Pr(|Y - \ell(X)| \leq w) \geq \frac{1}{2} \}$$

Reading the picture

- The true line needs positive width because of noise.
- A line through a point-mass cluster only needs a little extra mass from the majority.
- As contamination $\rightarrow \frac{1}{2}$, that width $\rightarrow 0$.
- Proved for population LMS only; LTS/MM are studied empirically.

The central shift

Instead of asking “Which line produces small residuals?”

RAShR asks “Which direction receives the strongest agreement from pairwise chords?”

Related work as one story: pairwise slopes → pairwise directions

Theil-Sen

median of all pairwise slopes
breakdown $\approx 29\%$

Repeated median

median of per-anchor medians
50% breakdown; rank effect

LMS / LTS

narrow strip / trimmed
residuals
coherent cluster can win

MM

robust start + M-step
efficient, inherits start

RANSAC

2-point models, count inliers
strong on some geometries

Hough transform

votes in parameter space
mode seeking

Shorth

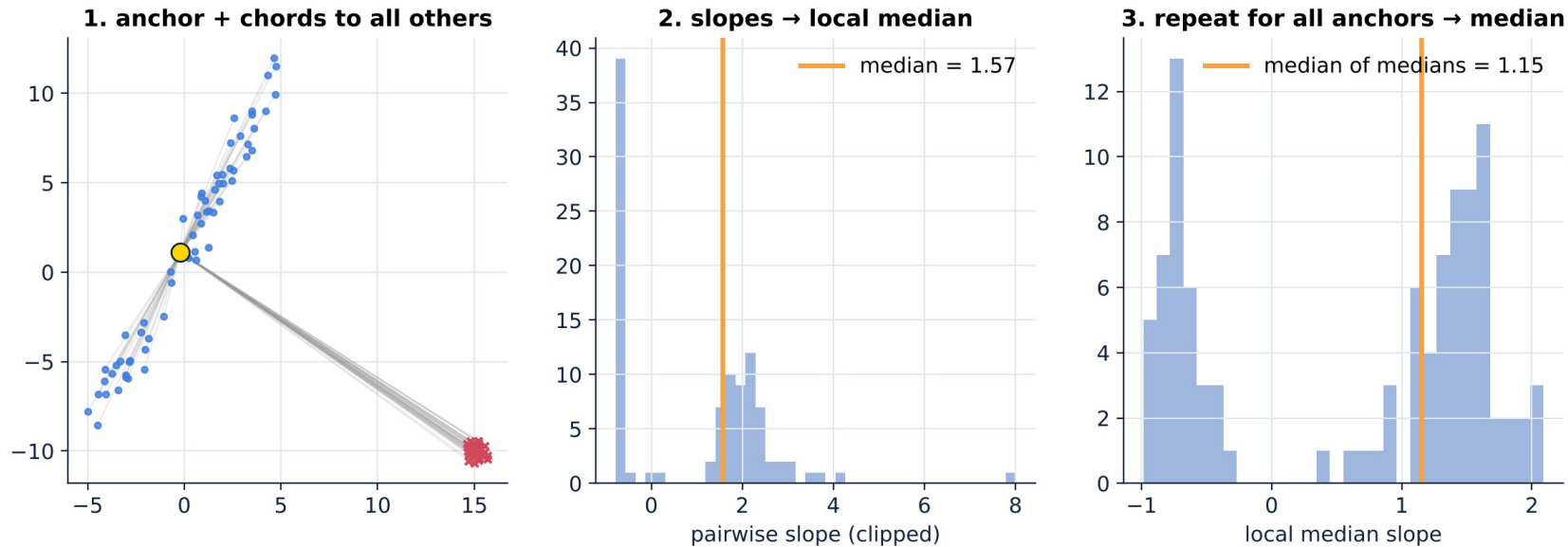
shortest half-interval
→ RAShR building block

Directional regression

orientation as geometry
→ same spirit

Each method: geometric object → what it measures → limitation → link to RAShR

Siegel's repeated median: median of local medians



1 anchor

›

2 chords

›

3 slopes

›

4 local median

›

5 all anchors

›

6 median of medians

$$\hat{\beta}_{\text{RM}} = \text{med}_i \text{med}_{j \neq i} \frac{y_j - y_i}{x_j - x_i}$$

Bridge to RAShR

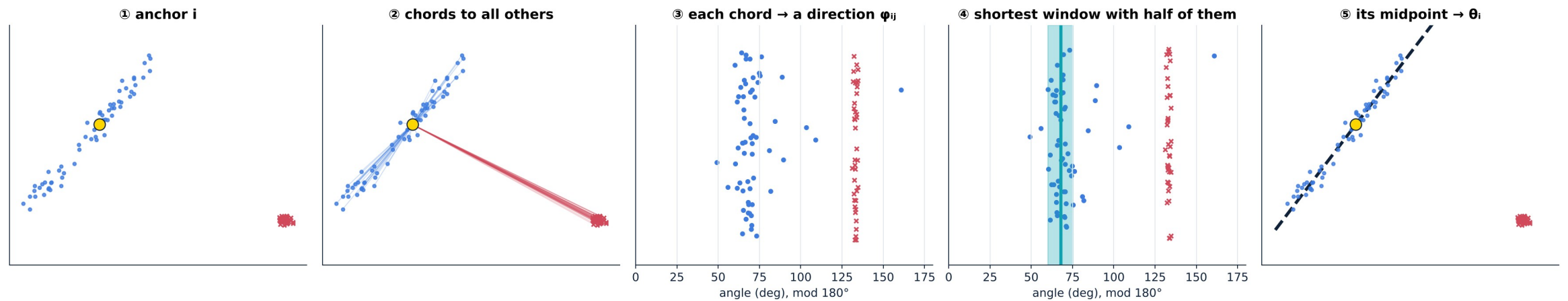
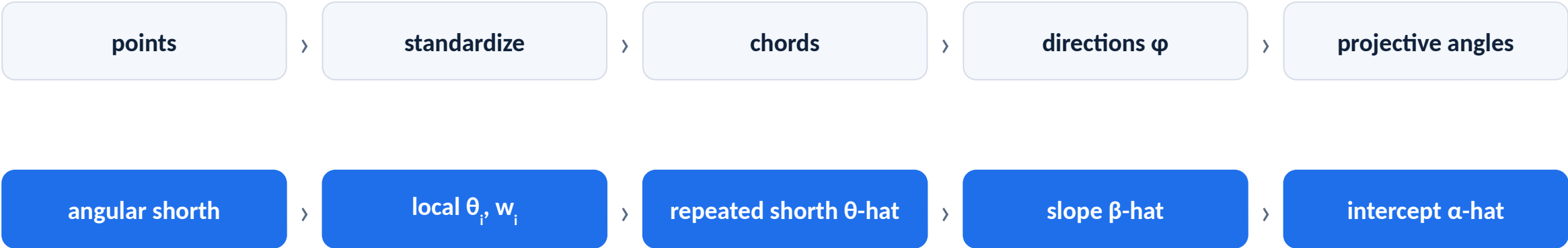
Keep the pairwise-geometric idea; replace slope/median aggregation by angular concentration through the shorth.

02

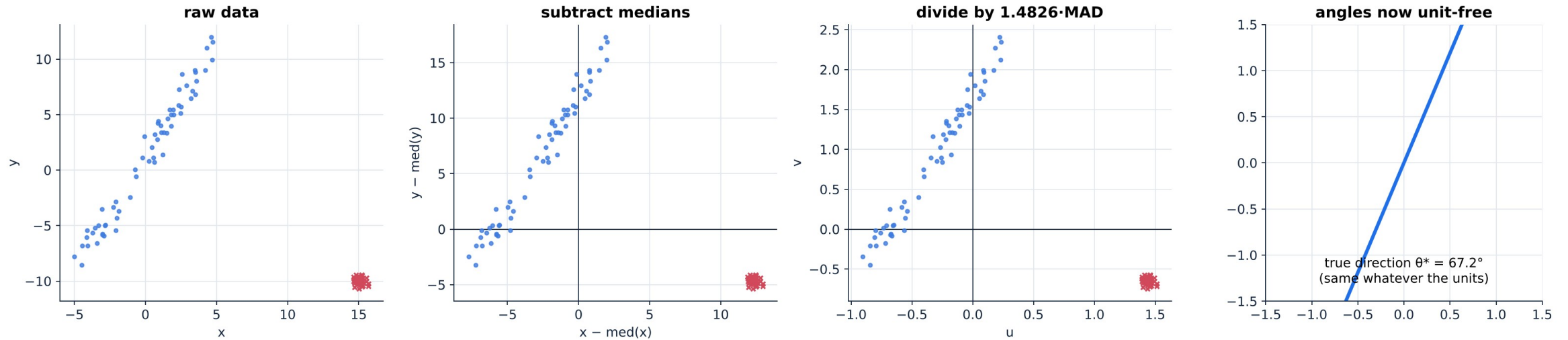
RAShR

points → chords → directions → projective angles → angular shorth → local directions → repeated shorth → slope → intercept → line

The whole path, in one picture



Step 1 — Angles depend on units, so standardize robustly



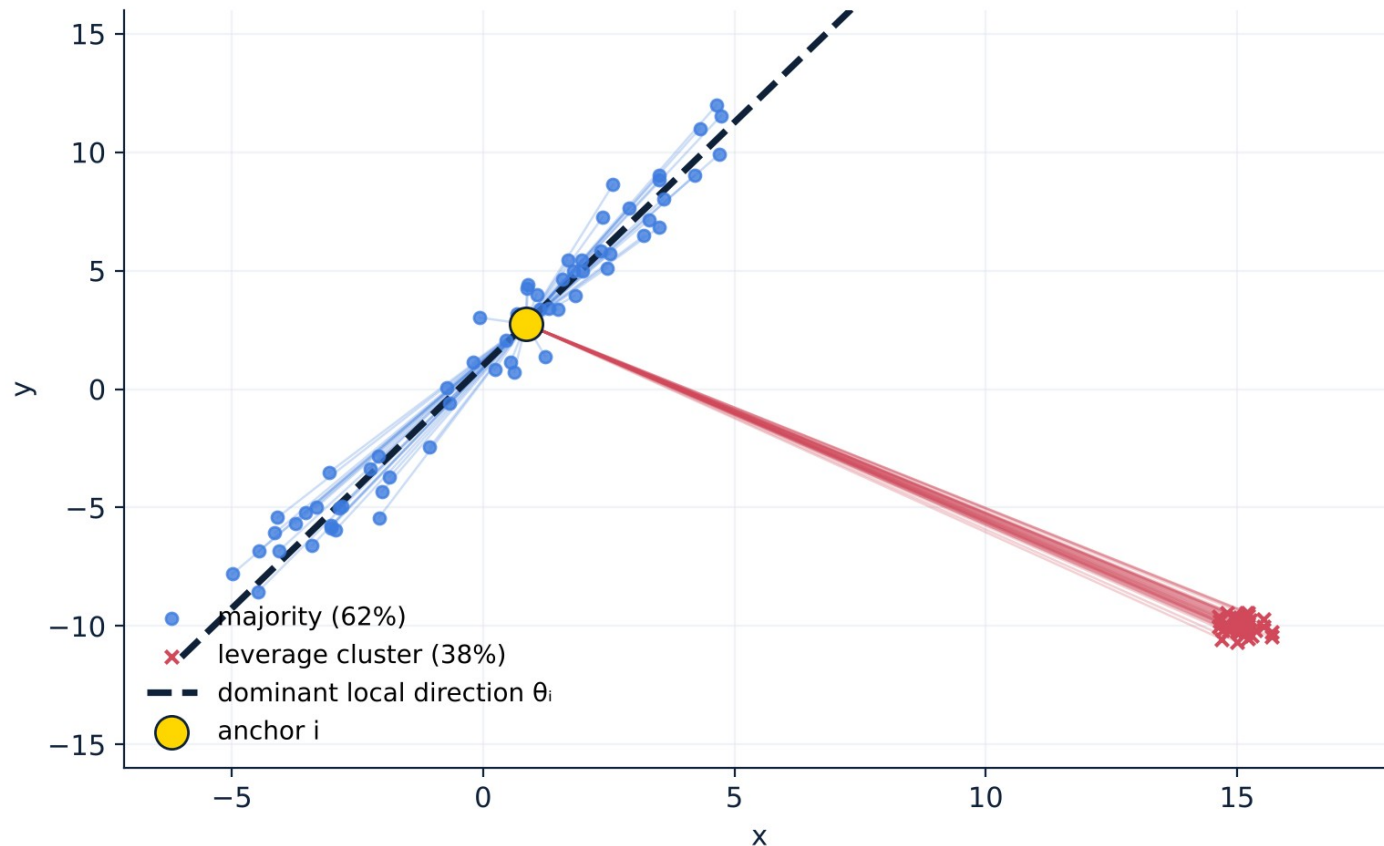
$$u_i = \frac{x_i - \text{med } x}{s_x}, \quad v_i = \frac{y_i - \text{med } y}{s_y}, \quad s = 1.4826 \cdot \text{MAD}$$

original data → robust centre → robust scale → standardized data

Details from the paper

1.4826·MAD = normal-consistent scale. If MAD = 0, use the standard deviation of that coordinate.

Step 2 — A chord is a vector; its direction is atan2

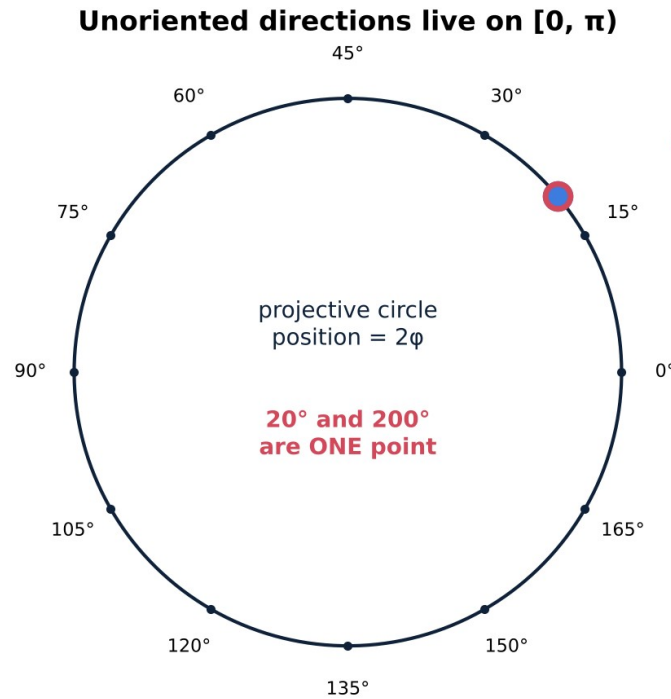


$$\varphi_{ij} = \text{atan2}(v_j - v_i, u_j - u_i) \bmod \pi$$

Why atan2?

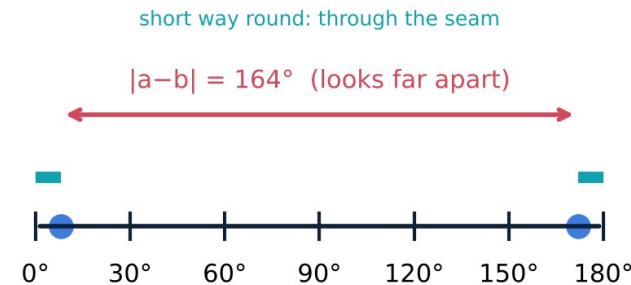
- It uses the vector $(\Delta u, \Delta v)$, not the ratio $\Delta v / \Delta u$.
- A nearly vertical chord ($\Delta u \approx 0$) is harmless.
- Majority-majority chords (blue) fan out near the true direction; chords to the cluster (red) all share one very different direction as seen from this anchor.

Step 3 — The projective circle: a line has no orientation



Wrap-around: a plain interval is not enough

$$d_{\pi}(a,b) = \min(|a-b|, 180^{\circ}-|a-b|) = 16^{\circ} \rightarrow \text{actually adjacent!}$$



0° and 180° are glued: the interval wraps around

$$d_{\pi}(a, b) = \min \{ |a - b|, \pi - |a - b| \}$$

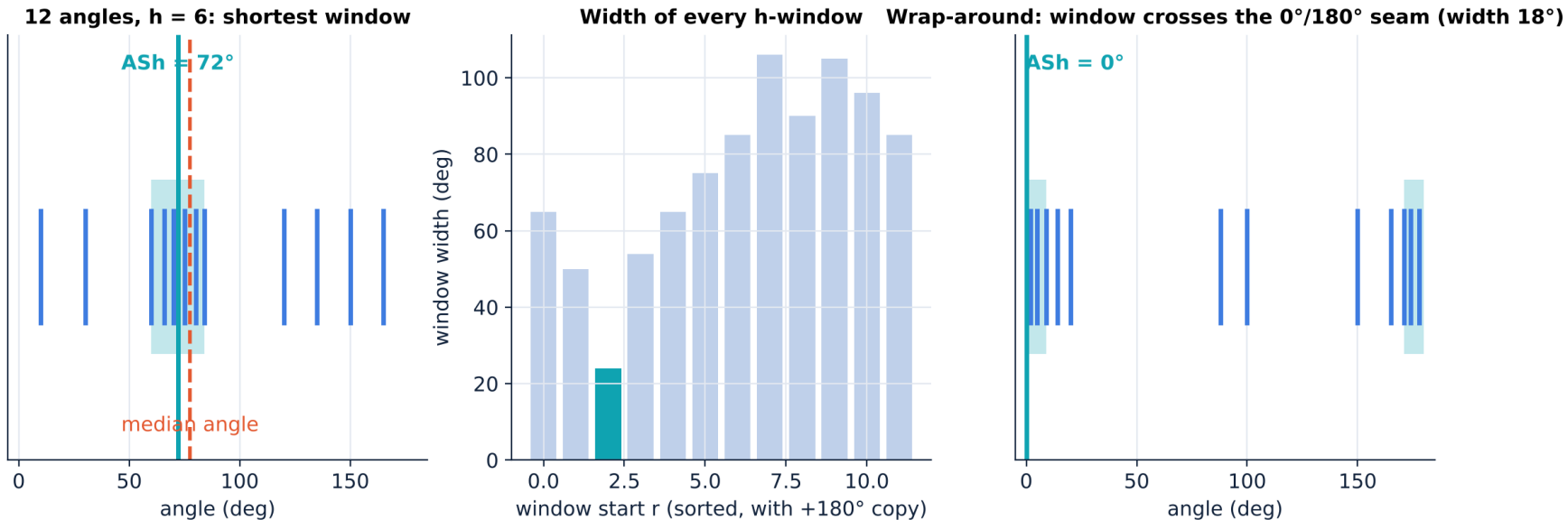
$$20^{\circ} \equiv 200^{\circ}$$

Opposite directions are the same unoriented line, so directions live on $[0, \pi)$ with 0 and π glued.

Wrap-around

5° and 175° are only 10° apart. An ordinary real interval cannot represent an arc crossing the seam.

Step 4 — Angular shorth: the shortest arc holding half the angles



on circle

sort

copy + π

slide h-windows

measure widths

shortest wins

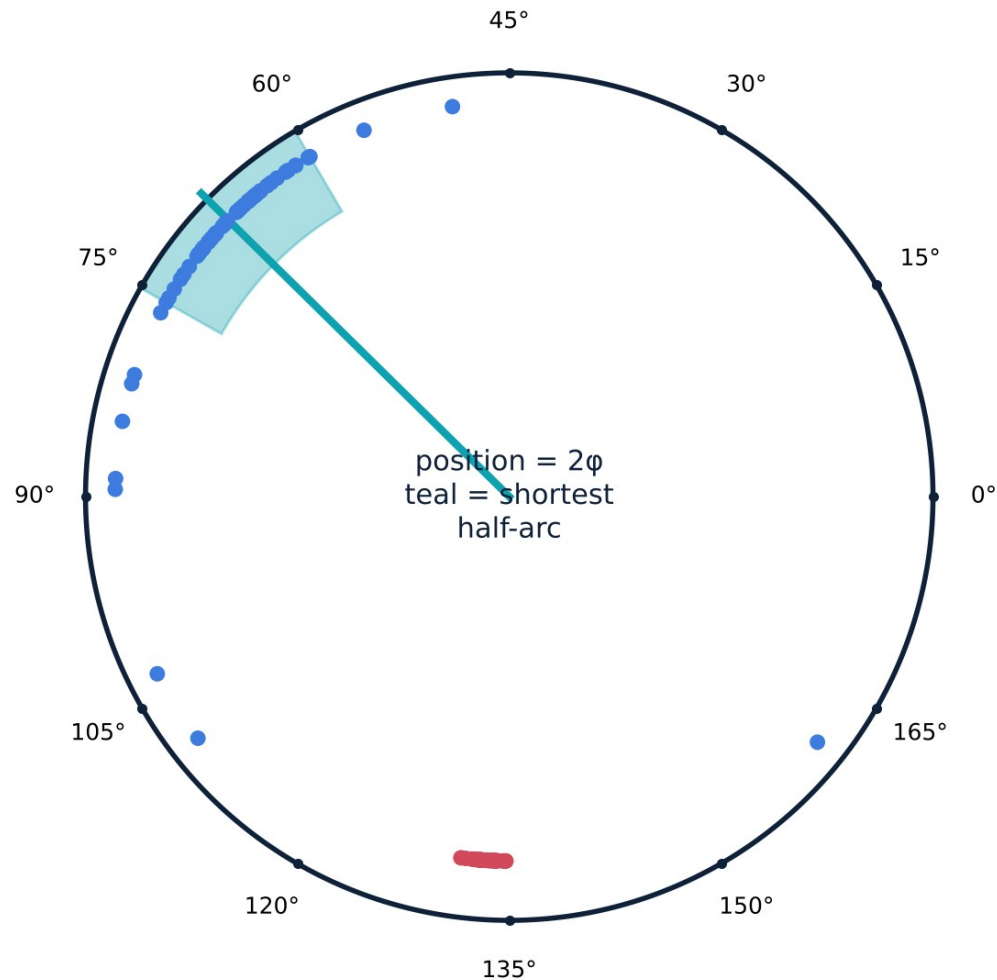
midpoint

$$\text{ASh}_h(A) = \frac{1}{2}(a_{(r^*)} + a_{(r^* + h - 1)}) \bmod \pi, \quad h = \lceil m/2 \rceil$$

Shorth ≠ median

Median angle = middle-ranked. Shorth = centre of the strongest concentrated half. Ties → smallest left endpoint.

Step 5 — Local direction: what one anchor sees



$$\Phi_i = \{\varphi_{ij} : j \neq i\}, \quad \theta_i = \text{ASh}(\Phi_i), \quad w_i = \text{width}$$

Small w_i

Strong directional agreement from anchor i .

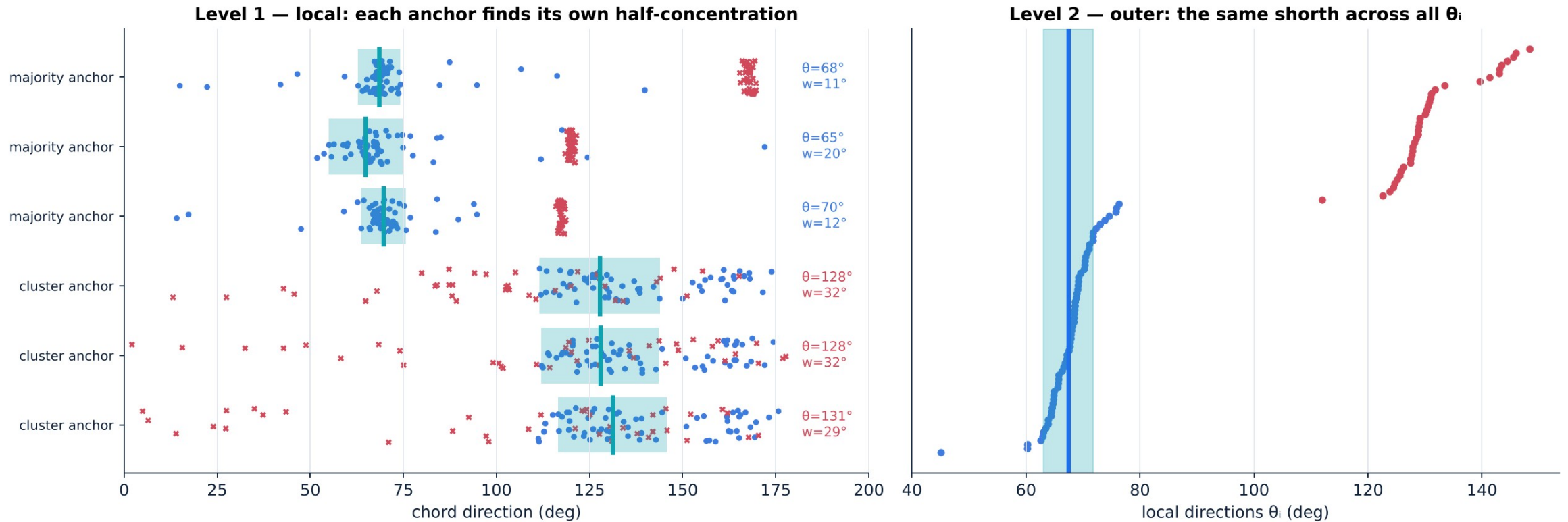
Large w_i

Weak agreement; no clear direction from this anchor.

On the circle

Blue dots: chords to majority points. Red dots: chords to the cluster. The teal arc is the shortest half-window; its midpoint is θ_i .

Step 6 — Repeated: the same shorth, twice



$$\hat{\theta} = \text{ASh}\{\theta_1, \dots, \theta_n\}$$

Local aggregation

inside each anchor's $n - 1$ chords ($h_L = \lceil (n-1)/2 \rceil$)

Outer aggregation

across the n local directions ($h_o = \lceil n/2 \rceil$)

Step 7 — Back to the original scale

 $\hat{\theta}$

>

 $\tan \hat{\theta}$

>

 $\times s_y$

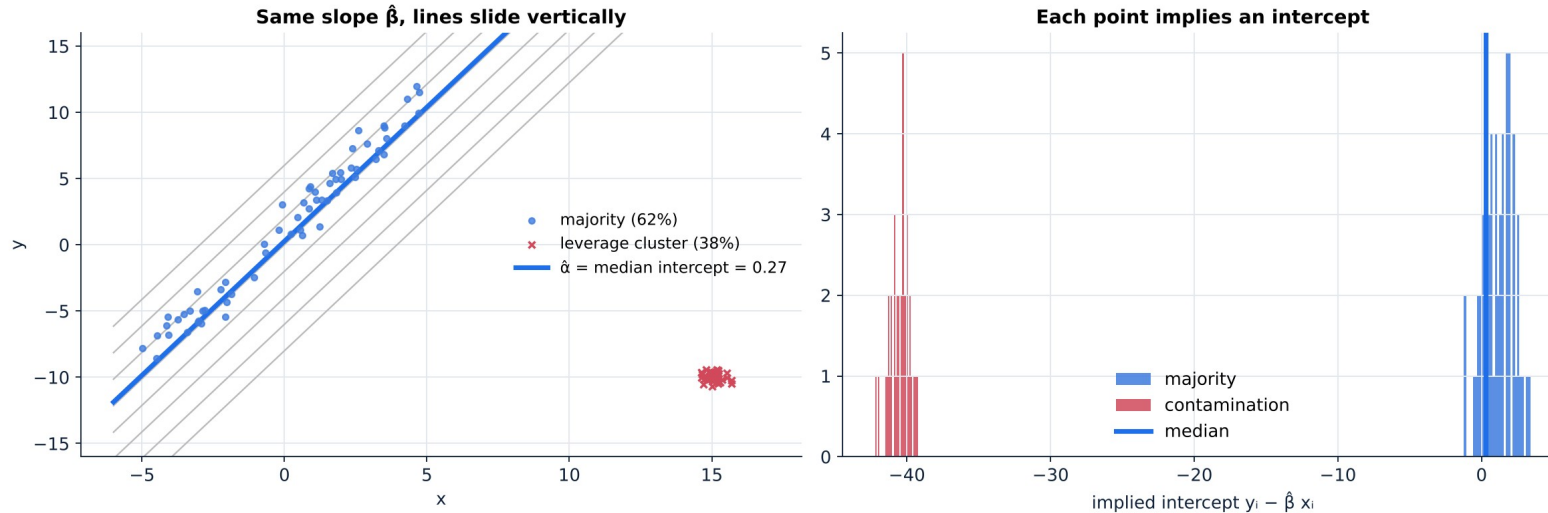
>

 $\div s_x$

>

 $\hat{\beta}$ (slope)

$$\hat{\beta} = \frac{s_y}{s_x} \tan \hat{\theta}, \quad \hat{\alpha} = \text{med}_i (y_i - \hat{\beta} x_i), \quad \hat{y} = \hat{\alpha} + \hat{\beta} x$$



Honest caveat

$\hat{\alpha}$ is a median over all observations. With 38% contamination at (15, -10) it can be pulled even when $\hat{\beta}$ is right; the paper evaluates the slope.

Algorithm and cost

raw observations

robust standardization

pairwise chord directions

local shortest half-window

local directions θ_i

second angular shorth

original-scale slope β -hat

median intercept α -hat

final line

$O(n^2 \log n)$

time: n anchors $\times$ sort $n-1$ angles

$O(n)$

extra memory, one anchor at a time

vs exhaustive subset search

Direct LMS/LTS subset search is $O(n^3)$ in the paper's implementations. Same order as a direct repeated median.

Deterministic

No random sampling. Ties go to the smallest left endpoint, so the same data always give the same line.

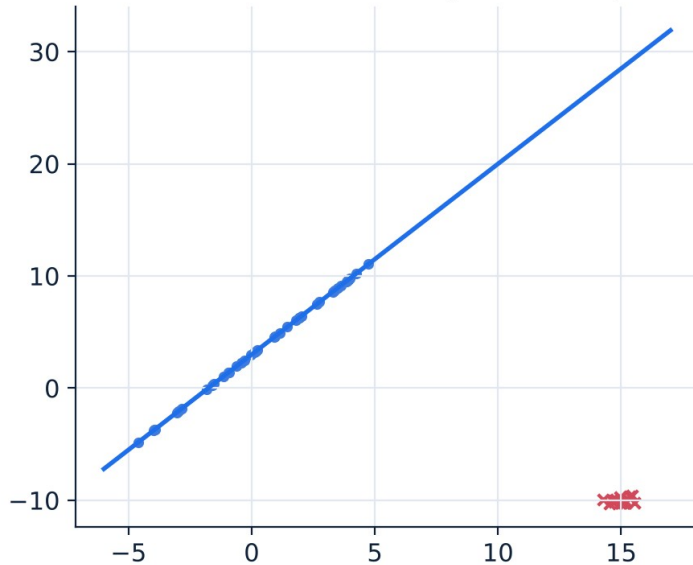
03

Theory through geometry

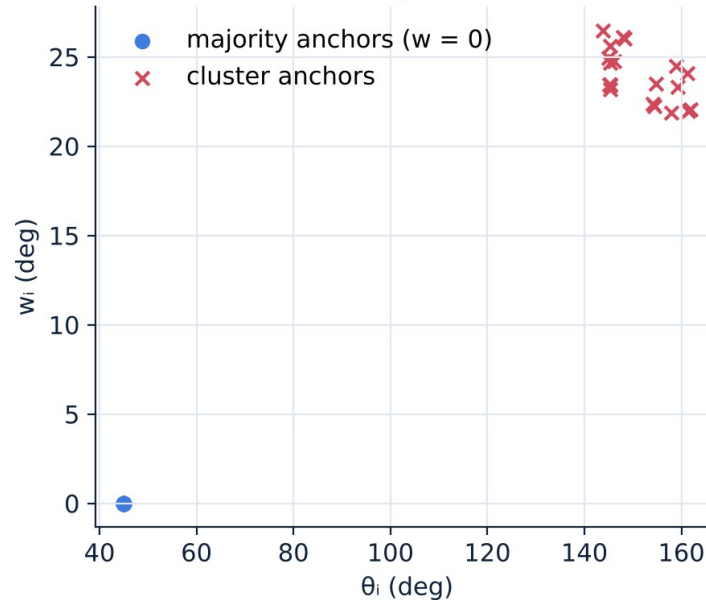
statement → meaning → geometry → experiment → significance

Proposition 1 — Exact fit (and Corollary 1)

Exact fit under contamination: $\hat{\beta} = 1.7000$, $\hat{\alpha} = 3.0000$



Zero-width majority directions



Statement

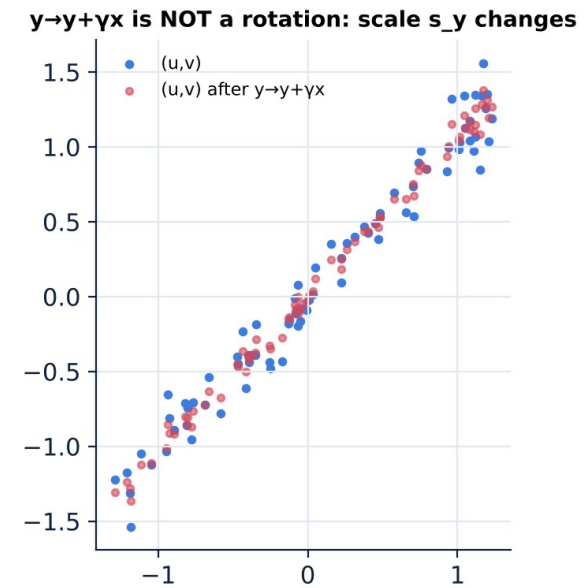
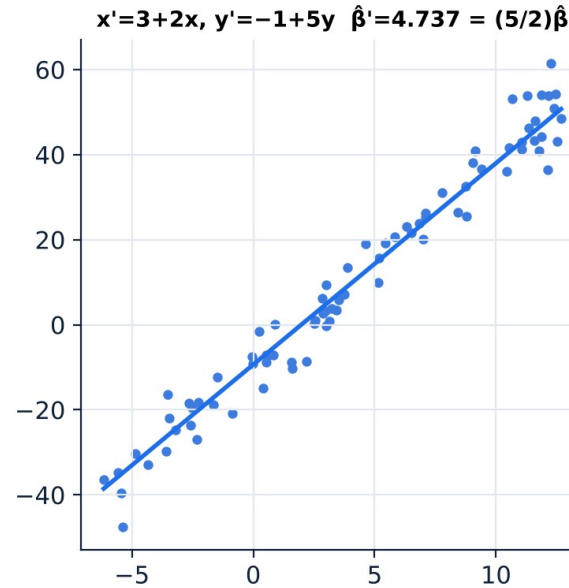
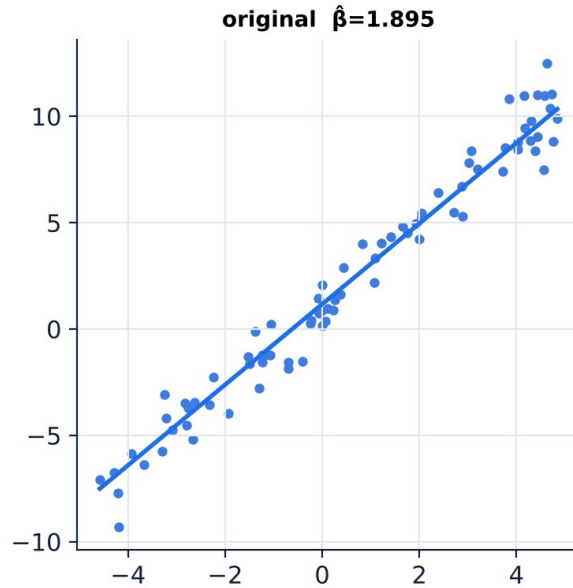
If $y_i = \alpha^* + \beta^* x_i$ with distinct x_i , then $(\hat{\alpha}, \hat{\beta}) = (\alpha^*, \beta^*)$.

All chords share one direction $\rightarrow$ all shorth widths are 0.

Corollary 1 — exact fit under contamination

If the majority lies exactly on a line, no contaminated point is on it, and $|B| < \lceil (n-1)/2 \rceil$, the line is recovered wherever the contamination is: a group smaller than the half-sample cannot form a competing zero-width window. (Shared with LMS and the repeated median.)

Proposition 2 — Translation and positive-scale equivariance



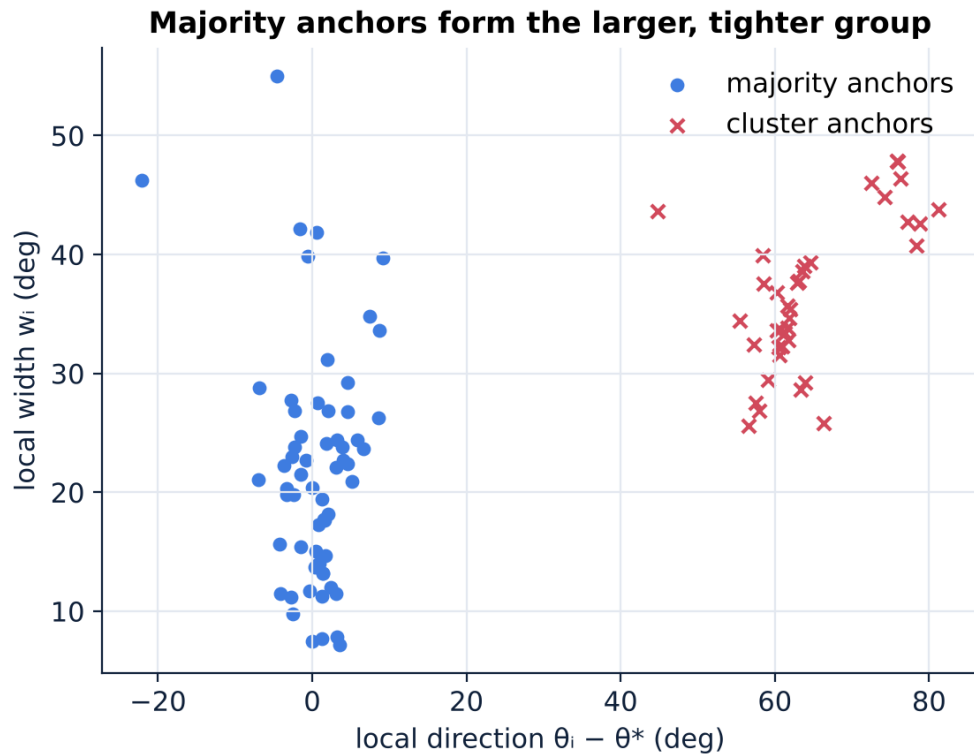
$$x' = a_x + b_x x, \quad y' = a_y + b_y y \quad (b_x, b_y > 0) \Rightarrow \hat{\beta}' = \frac{b_y}{b_x} \hat{\beta}$$

Standardization removes translation and positive rescaling, so the same chords and shorths are selected.

Limitation

Not fully affine equivariant: $y \rightarrow y + \gamma x$ changes s_y , so it is not a simple rotation of the standardized data.

Separation: lemma, local condition, outer condition



Lemma 1

If every arc with $\geq h$ elements that touches B is strictly wider than A's shortest arc, then $\text{ASh}(A \cup B) = \text{ASh}(A)$.

(L) Local

At each majority anchor, majority-majority chords form a narrower half-window than any involving contamination.

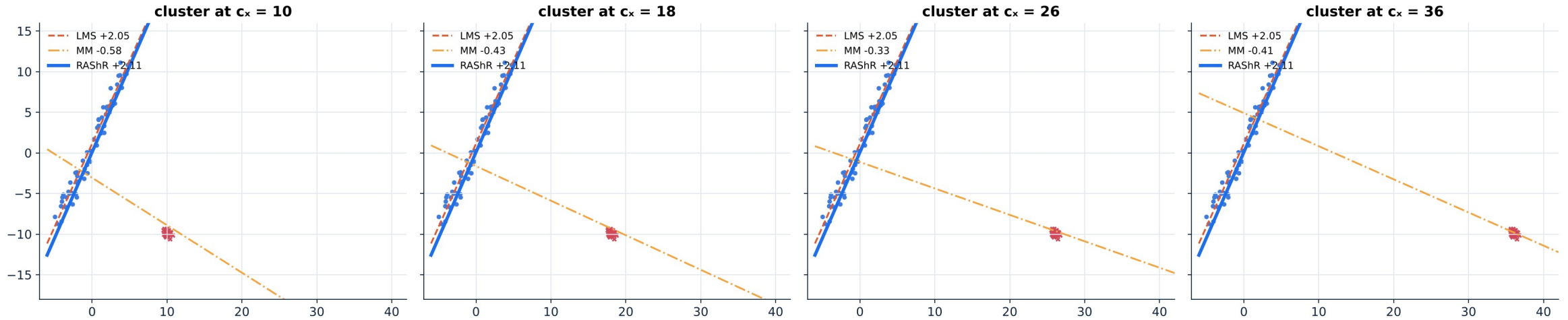
(O) Outer

Majority-anchor directions form a narrower half-sample window than any involving contaminated anchors.

Size

$|G| > n/2 \Rightarrow |G| - 1 \geq h_L$ and $|G| \geq h_O$:
both stages can be built from majority points alone.

Theorem 1 — Zero influence of separated contamination



$$\hat{\theta} = \text{ASh}_{h_O} \{ \text{ASh}_{h_L}(\Phi_i^G) : i \in G \}$$

Paper: slope exactly 2.058 once the cluster is far enough (its sample). Ours differs (different sample); the constancy is the point.

What it says

Once (L), (O) and $|G| > n/2$ hold, moving, adding or removing contaminated points does not change $\hat{\beta}$.

Caveat: the standardization medians/MADs must stay unchanged — contamination can still affect Eq. (2).

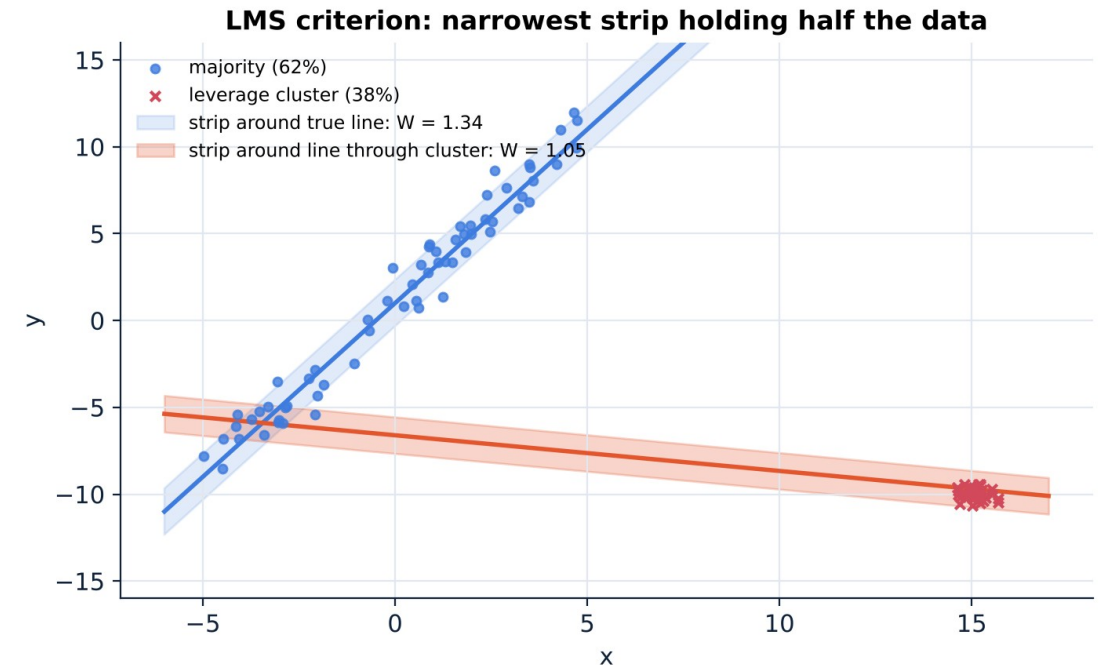
Proposition 3 — Why coherent contamination can attract LMS

Model

$(X, Y) \sim (1-\varepsilon)P_G + \varepsilon \delta_c$, with $P_G: Y = \alpha^* + \beta^*X + e$, $e \sim N(0, \sigma^2)$. The point mass c is at vertical distance $\Delta > 0$ from the majority line.

Statement

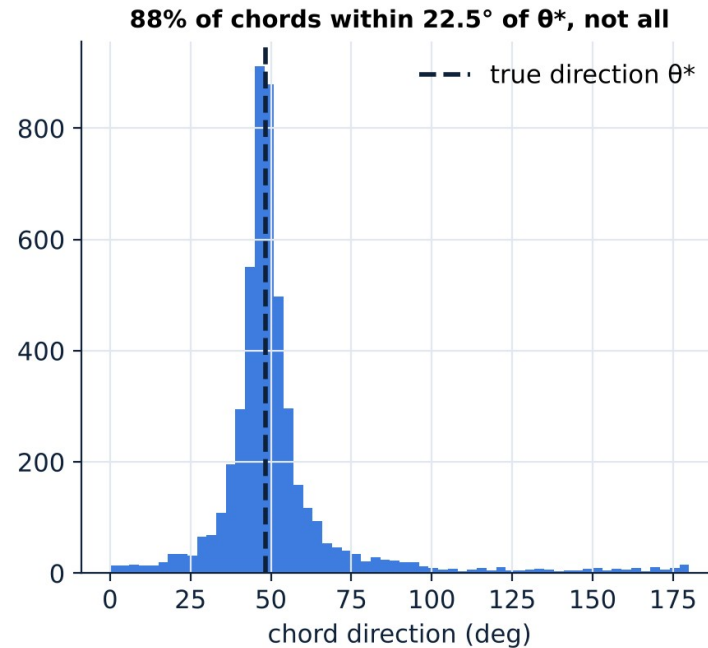
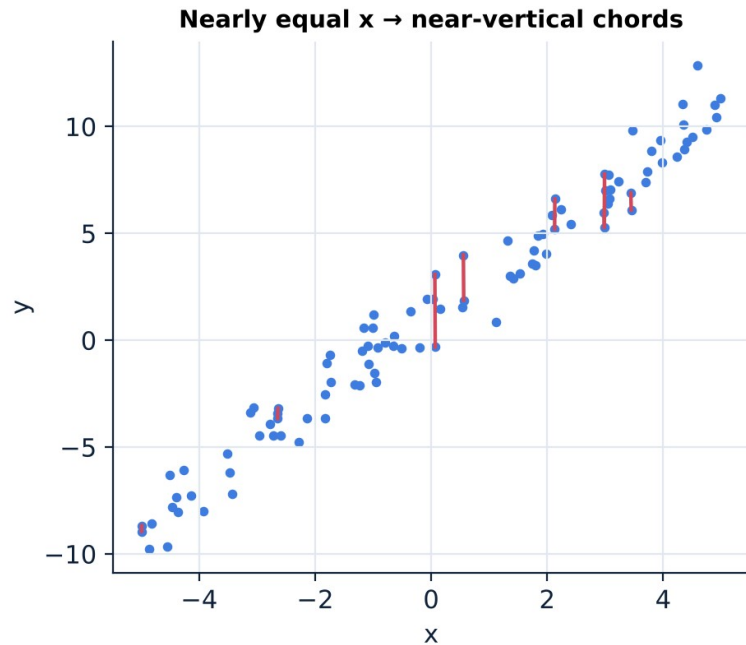
There is $\varepsilon_0 < \frac{1}{2}$ such that for $\varepsilon \in (\varepsilon_0, \frac{1}{2})$ every population LMS solution ℓ has $|c_y - \ell(c_x)| \leq W(\ell)$ with $W(\ell) \rightarrow 0$ as $\varepsilon \uparrow \frac{1}{2}$. The majority line is not an LMS solution.



Scope

Proved for population LMS, asymptotically in $\varepsilon \uparrow \frac{1}{2}$. For LTS and MM only empirical evidence (Section 5), no theorem.

Remark — why “all chords near θ^* ” is too strong



The problem

Two points with almost equal x and a little vertical noise give a chord with an almost arbitrary direction. Requiring every majority chord within $\delta < \pi/8$ of θ^* is far too restrictive for noisy data.

What RAShR needs instead

Concentration of a half-sample of chords — exactly what the shorth detects — not concentration of all of them.

04

Simulation evidence

where RAShR helps — and where it does not

Simulation design (paper, Section 5)

Data

$n = 100$, $x \sim U[-5, 5]$, $y = 1 + 2x + e$, $e \sim N(0, 1)$

A fraction f is replaced by:

- ① compact leverage cluster at $(15, -10)$, spread 0.3
- ② competing line $y = 5 - 1.25x + e_c$, $e_c \sim N(0, 0.5^2)$
- ③ funnel: majority sd $0.2 + 0.6|x|$ plus a line with sd 0.3

Metric $|\hat{\beta} - 2|$; failure = $|\hat{\beta} - 2| > 0.5$; 60 replications.

Competitors

LMS · LTS (10 best elemental fits + 20 C-steps) · MM (LTS start, MAD scale, Tukey bisquare) · RANSAC (2-point samples, MAD-based threshold, 1000 trials)

Reproducibility note

All curves in this deck come from our re-implementation with fixed seeds. Some constants (e.g. RANSAC's threshold) are not fully specified, so we compare patterns; the paper's own numbers are shown separately.

Limitation first: RAShR is not universally better

Paper — median $|\hat{\beta} - 2|$

setting	RAShR	MM	RANSAC
Clean data	0.033	0.024	0.022
20% vertical contamination	0.040	0.027	0.037
10% bad leverage	0.041	0.020	0.021

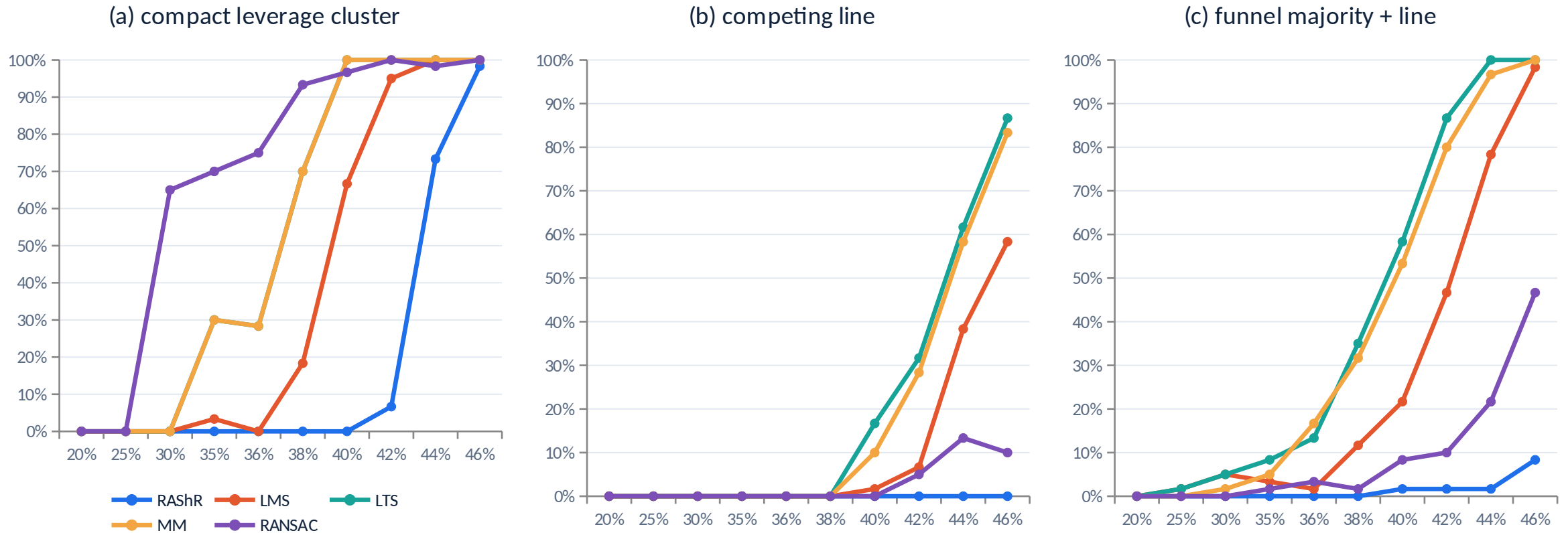
Take-away

At low contamination or in standard settings MM is generally the more efficient choice. RAShR's advantage needs a contaminating group that is both large and geometrically coherent.

This project's re-implementation (100 reps)

setting	RAShR	LMS	LTS	MM	RANSAC
Clean	0.043	0.054	0.051	0.022	0.022
20% vertical	0.034	0.052	0.056	0.023	0.056
10% bad leverage	0.042	0.054	0.067	0.022	0.022
20% competing line	0.045	0.070	0.060	0.032	0.061

Failure rate vs contamination (our re-implementation)



Failure = $|\hat{\beta} - 2| > 0.5$. Below $\approx 30\%$ the high-breakdown methods behave alike; between $\approx 35\%$ and 46% they separate. Compact cluster at 46% : every method fails.

What the paper reports (Figure 2 text)

(a) Compact leverage

36%: RAShR 0 · LMS 2 · LTS/MM 23 · RANSAC 50

40%: RAShR 0 · LMS 58 · LTS/MM 100 · RANSAC 77

44%: RAShR succeeds in 40% of samples; every competitor fails in $\geq 97\%$

46%: all methods fail

(b) Competing line

RAShR: no failures through 46%

At 46%: LMS 60 · LTS/MM 93 · RANSAC 12 · RAShR 0

RANSAC remains a strong competitor on this geometry

(c) Funnel + line

40%: RAShR 0 · LMS 25 · LTS 62 · MM 60 · RANSAC 5

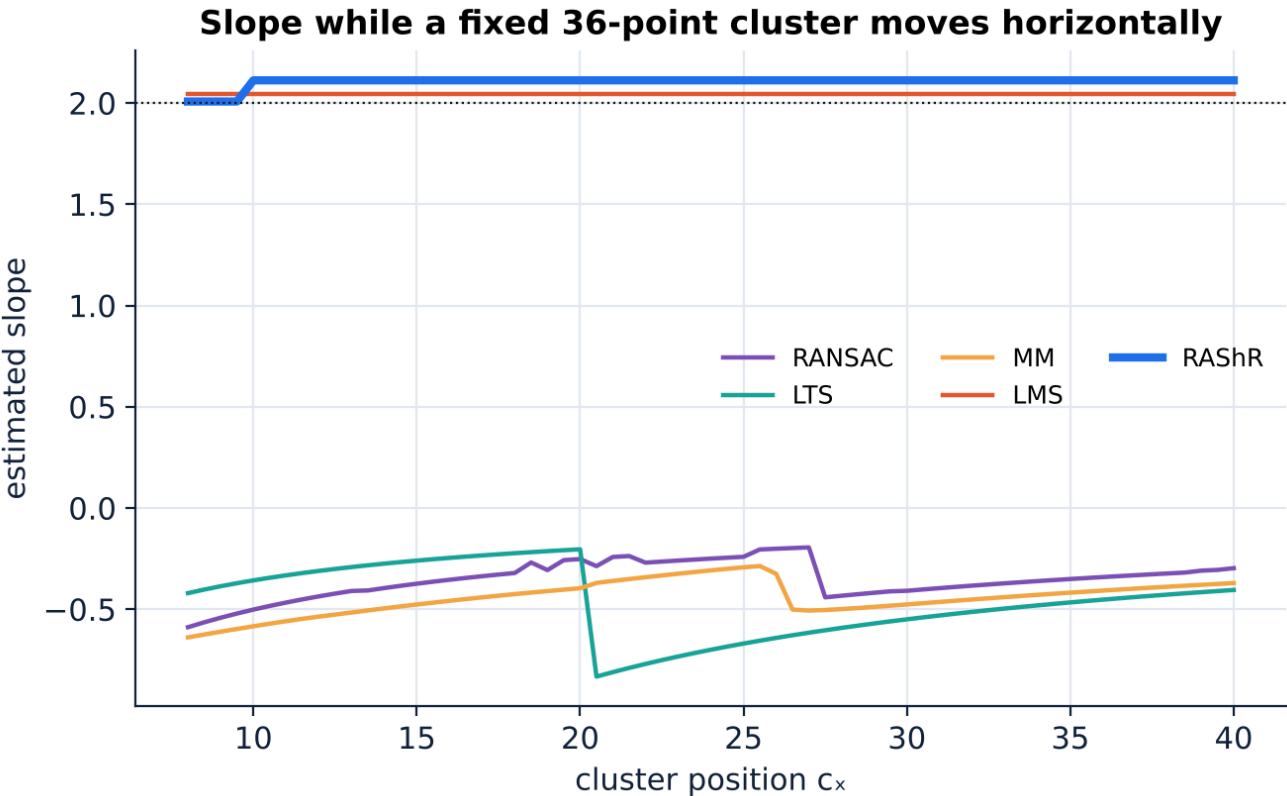
44%: RAShR 3 · LMS 83 · LTS 100 · MM 98 · RANSAC 40

46%: RAShR 8

Not a general “near-50%” guarantee

A finite regime ($\approx 35\text{--}46\%$) in which RAShR can stay near the majority line after several established methods have begun to follow the competing structure. All numbers: failure rate in %.

Figure 3 geometry: a moving cluster, and the 2.058 plateau



2.058

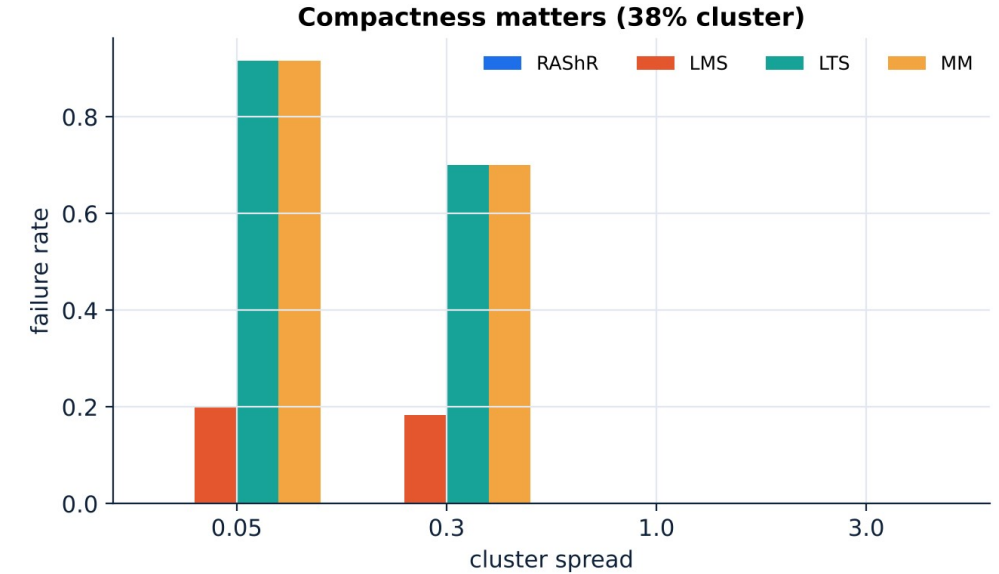
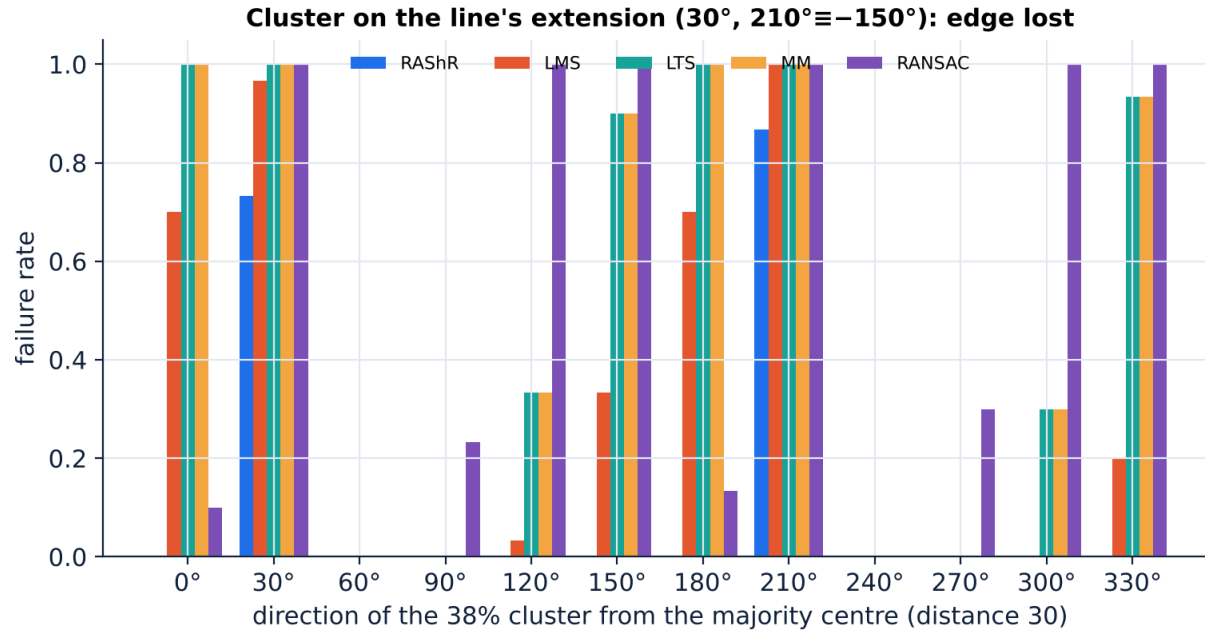
RAShR slope once the cluster is far enough (paper's sample)

Reading it

RAShR's slope becomes exactly constant — consistent with Theorem 1. Residual-based fits (LTS, MM, RANSAC) jump to the cluster when it is closer; each has a different switching point.

Our sample's plateau value differs from 2.058 (different majority sample); constancy is the point.

When the advantage disappears



Direction $\approx 30^\circ$ / -150°

Cluster on the extension of the majority line. $-150^\circ \equiv 30^\circ$ on the projective circle, so chord directions overlap and separation is lost (paper: 78% and 75% failure at distance 30).

Compactness

Spread 0.05: RAShR 0 failures, LMS 25%, LTS/MM $\approx 98\%$ (paper). Spread 1 or 3: all fail $\leq 2\%$ — the cluster is no longer a compact coherent group.

When is RAShR useful?

Use it when

≈ 35–46% of the data form a compact, coherent structure that is angularly separated from the majority line, and a deterministic estimator is wanted.

Prefer MM / RANSAC when

Data are clean or lightly contaminated, contamination is diffuse or arbitrary, or efficiency matters most.

Known limits

Cluster on the line's extension ($\approx 30^\circ$, -150°); compact cluster at 46%: all fail; not fully affine equivariant; scales and intercept use all data.

direction, not residual → concentrate twice → robust slope